

Sebastián Eduardo Puentes Prieto

AI Systems Architect | LLM/RAG Platform Engineer | Backend Engineering Lead

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PROFESSIONAL PROFILE

AI Systems Architect and Backend Engineering Leader with 15+ years of experience building distributed systems and production-grade LLM platforms. Expert in multi-agent orchestration (LangGraph), graph-augmented RAG architectures, neuro-symbolic pipelines, and MCP-native tooling. Deep proficiency in structured LLM generation (DSPy, instructor), knowledge graph construction (Neo4j), and multi-store context assembly across vector, graph, and document stores. Proven ability to design and lead agentic development workflows — directing coordinated LLM agent teams to deliver complete, production-ready platforms. Broad backend expertise across Python, Go, Ruby, Node.js, and AWS ecosystems, recognized for building systems that balance architectural rigor with delivery velocity.

PROFESSIONAL EXPERIENCE

InAdvance Consulting Group | *Agentic Development Lead* — 2025–Present

PII Sentinel — Neuro-Symbolic Compliance Platform

- Designed and delivered PII Sentinel, a neuro-symbolic compliance automation platform combining transformer-based NER (Presidio), LLM-driven ambiguity resolution, and a graph-augmented RAG layer (Neo4j compliance knowledge graph + DPV ontology grounding) to automate PII discovery, legal obligation mapping, and remediation reporting across PostgreSQL, MySQL, filesystem, and SSH-backed data sources.
- Full LangGraph-orchestrated scan pipeline — inventory ingestion, multi-signal PII classification, LLM ambiguity arbitration, compliance graph traversal, dual-narrative report generation — delivering **full DPIA inventory for DPOs**
- Stack delivered through agent-directed development: FastAPI backend, React frontend, Neo4j knowledge graph, scan connectors, PDF reporting, and Rule Forge HITL curation workflow; 6 control frameworks ingested (NIST/ISO/CIS/PCI-DSS/SOC2/COBIT).

BluePrism VBO → N8N — LLM-Assisted Automation Migration

- Designed and implemented an LLM-assisted migration pipeline converting BluePrism Visual Business Objects (VBOs) into N8N workflow definitions; structured LLM generation semantically interprets VBO XML automation logic and produces idiomatic N8N JSON configurations, enabling large-scale decommissioning of legacy RPA infrastructure.

Agile Engineering Leadership

- Led traditional agile teams on implementation and continuous improvement projects for financial-sector clients, managing sprints, resources, and delivery using Java and Go frameworks.

Andes Path LLC | *Senior Software Engineer* — 2025

- Executed the complete refactoring of a monolithic system to a microservices architecture, improving scalability and facilitating future maintenance.
- Designed and implemented CI/CD pipelines from scratch in GitHub Actions to automate the development lifecycle.

Checkr | *Senior Backend Engineer* — 2023–2025

- Led the migration of legacy Sidekiq-based systems to an event-processing ecosystem with Kafka, improving resilience and throughput by over 30%.
- Designed and developed high-traffic RESTful APIs in Ruby on Rails for the business core, managing over 500k daily requests with low latency.
- Created AI Assistant agents using Credal and internal data sources, both pre- and post-MCP.
- Contributed to full-stack development of the main platform using Ruby on Rails and ReactJS, delivering key features for back-office users.

Fulcrum | *Senior Backend Engineer* — 2022–2023

- Developed geospatial microservices in AWS using AppSync (GraphQL), Lambda, and ArcGIS, processing large volumes of geographic data in real-time.
- Designed and developed high-traffic RESTful APIs in Ruby on Rails for core and back-office management.

Moovx | *Senior Backend Engineer* — 2022

- Designed the backend architecture for a high-performance e-commerce platform using Go, Rust, and Spring Boot, focused on scalability and low latency.
- Implemented a 100% serverless architecture in AWS, optimizing operational costs.
- Created and maintained internal libraries published in GitHub Packages to standardize development across the organization.

Globant | *Senior Backend Developer* — 2020–2022

- Developed microservices in Go, Node.js (NestJS), and Java (Spring Webflux) for top-tier clients, following Domain-Driven Design patterns.
- Created conversational applications for Alexa and Google Assistant with the Jovo framework, integrating AI and NLP services.
- Developed voice interfaces with AI assistants using TensorFlow and Dialogflow.

Everis Chile | *Software Architect / IT Consultant* — 2015–2020

- Led the architecture of web applications (AngularJS, Node.js, Microservices) for key clients in the banking and government sectors.
- Developed custom Cordova plugins and designed adapters to integrate legacy SOAP systems.
- Participated in the modernization of J2EE projects for the public sector.

KEY PROJECTS & ADDITIONAL EXPERIENCE

LAIN-mcp — **Codebase Knowledge Graph MCP Server** (*Personal Project, 2026*)

<https://github.com/spuentesp/lain>

- Built LAIN-mcp, a high-performance MCP server in Rust exposing a persistent petgraph-based knowledge graph of a codebase via 21 MCP tools to AI coding agents. Provides macroscopic architectural awareness — blast radius analysis, co-change detection, cross-file dependency traversal — well beyond the reach of standard RAG or LSP approaches.
- Implemented local ONNX-based semantic code search (no API keys, no data egress), sub-millisecond graph traversal, and git co-change analysis to surface files that evolve together.
- Stack: Rust, petgraph, ONNX (all-MiniLM-L6-v2), git2, MCP protocol (stdio/HTTP).

MONITOR — **Narrative Intelligence System** (*Personal Project, 2025–Present*)

https://github.com/spuentesp/monitor_dm_system

- Designed and built a multi-agent, graph-augmented RAG system combining Neo4j (canonical truth graph), MongoDB (narrative memory), and Qdrant (semantic recall) into a unified retrieval layer with provenance-aware canonization.
- 9 specialized LLM agents orchestrated by 5 LangGraph state machines with MongoDB-checkpointed durability, DSPy prompt modules, and instructor-enforced structured outputs.
- Full document ingestion pipeline (PDF → chunking → embedding → structured knowledge extraction) producing reusable knowledge packs with source profiling and conflict-aware world application.
- Stack: Python, FastAPI, Next.js, Neo4j, MongoDB, Qdrant, Redis, LangGraph, DSPy, Anthropic/OpenAI APIs, Docker.

Additional:

- Senior Developer (SHIFT): Backend for a personnel management app in Node.js (NestJS) and frontend in Ionic 4, integrating RESTful and GraphQL APIs.
- Mobile Consultant (Amaris): Mobile applications with React Native and backend services in Node.js.

- Independent: Published 'Apocalipsis Zombie Chile' on Google Play — Ionic 3 with AR (A-Frame) and Node.js backend with Express, JWT, and WebSockets.

KEY TECHNOLOGIES & SKILLS

AI/RAG Architecture: Multi-agent orchestration (LangGraph), graph-augmented retrieval, multi-store context assembly (Neo4j + Qdrant + MongoDB), structured LLM generation (DSPy, instructor), neuro-symbolic pipelines, MCP tooling & server design, vector search (Qdrant), semantic chunking, knowledge graph construction, Presidio NER, Anthropic/OpenAI APIs

Languages: Python, Go, Ruby, TypeScript/JavaScript, Rust, Java, SQL

Backend Frameworks: FastAPI, Gin/Echo (Go), Ruby on Rails, Spring Boot, NestJS, Express

Frontend: React.js, Next.js, Angular, Ionic, React Native

Cloud & DevOps (AWS): Lambda, API Gateway, AppSync, DynamoDB, S3, EC2, ECS, SNS, SQS | Docker, GitHub Actions, GitLab CI

Databases: Neo4j, PostgreSQL, MongoDB, Qdrant, Redis, MySQL, DynamoDB

Architecture & Patterns: Microservices, Serverless, Domain-Driven Design, Event-Driven Architecture, Neuro-Symbolic Pipelines

Messaging & Streaming: Kafka, RabbitMQ, Sidekiq

Protocols & APIs: REST, GraphQL, gRPC, SOAP, Protobuf, MCP

EDUCATION & CERTIFICATIONS

Magíster en Ingeniería Informática

Universidad de La Frontera, Temuco, Chile (Thesis defense stage, 2024–2026). Research focus: velocity-weighted inventory optimization for micro-retailers using asymptotic weighting functions. Explores mathematical models for demand-driven stock allocation in resource-constrained small businesses, with empirical validation against real retail data.

Ingeniero en Informática (Software Engineer)

Universidad Tecnológica de Chile INACAP, Santiago (2014). Five-year professional engineering degree covering software architecture, systems design, databases, and project management.

Analista Programador (Programmer Analyst)

INACAP, Santiago (2011). Three-year technical degree in software development, algorithms, and applied computing.

Certifications: TOEIC 945/1000 — Advanced Level | Scrum Foundation Professional Certificate (SFPC)